

**Adopted Levels: not observed**

$S(n)=24210$  syst;  $S(p)=-60$  syst;  $Q(\alpha)=-9610$  syst 2021Wa16

$\Delta S(n)=500$ ,  $\Delta S(p)=360$ ,  $\Delta Q(\alpha)=350$  (syst, 2021Wa16).

$S(2p)=-2510$  300,  $Q(\varepsilon)=16110$  360,  $Q(\varepsilon p)=16990$  300 (syst, 2021Wa16).

Isotope discovery: none.

$S(2p)$  is negative and  $S(p)$  may be consistent with zero.  $^{34}\text{Ca}$  is a candidate for ground state two-proton emitter.

Two-proton decay calculations: 2024Li59, 2024Ji17, 2023Me08, 2022Ro15, 2021Li27, 2020Ma35, 2020Cu01, 2017Sa70, 2004Pf02, 2003Gr04, 2005Pf01, 2003Gr24, 2001Gr29.

Structure calculations: 2002Zh09, 2001Yo12, 1998Co30, 2005Wa15, 2005Zh28, 1997Co19, 1997Pa38, 1996Sh20, 1995La03.

Giant-resonance calculations: 2021Li26, 2001Sa57, 1997Ha57, 2003Ma71, 2001Sa77, 1999Sa22, 1998Ha09, 1998Ha39, 1998Sa28, 1997Ha55.

 $^{34}\text{Ca}$  Levels

| <u>E(level)</u> | <u>J<sup><math>\pi</math></sup></u> | <u>T<sub>1/2</sub></u> | <u>Comments</u>                                                                                                                                   |
|-----------------|-------------------------------------|------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------|
| 0               | 0 <sup>+</sup>                      | <35 ns                 | %p=?; %2p=?<br>T <sub>1/2</sub> : upper limit estimated from expected cross sections of $^{40}\text{Ca}$ fragmentation at LISE, GANIL (1996PoZZ). |